

UCLA TA1 — AIQ Phase 1 Submission Summary

UCLA TA1 Team (VeriStressGT)

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Submission. Evaluation card `cards/thrust1_soundness.yaml`, branch `evaluation-card` of <https://github.com/dtroxell19/VeriStressGT>. Setup and run instructions are in `AIQ_README.md` in the repository.

1 Approach

A neural-network verifier either proves that a network is robust, meaning no small perturbation of an input can change its output (UNSAT), or finds a perturbed input that does (a counterexample, SAT). Verifiers are used to certify networks in safety-critical settings, so a verifier bug that wrongly reports “robust” gives false assurance. Catching such bugs requires knowing the right answer, which is what VeriStressGT provides.

VeriStressGT is a benchmark instance generator that tests neural-network verifiers on networks whose robustness answer is known by construction, so a wrong verdict is provably wrong. Existing benchmarks have no ground truth and fall back on majority vote between verifiers.

- **Default Benchmark:** 50 instances (see section Scalability), pre-built in the repository. 24 are UNSAT (i.e. robust) by analytic construction. 26 are SAT (non-robust), each with a counterexample checked in float64 and onnxruntime float32.
- **Verifier pool:** 2 sound in-house controls; 5 real verifiers (α - β -CROWN, PyRAT, nenum, NeuralSAT, Marabou); and 42 deliberately buggy verifiers: 12 mutants of our reference verifier plus 6 output-level faults on each real verifier.
- **Scoring:** every definitive verdict is judged against the known label. The same verdicts are also judged by majority vote, as a baseline.

2 Impact and Comparison with Other Benchmarks

VeriStressGT exposes 39 of the 42 planted bugs; existing benchmarks expose at most 6. We ran the same verifier pool on two VNN-COMP benchmarks, `mnist_fc` (30 of its 90 instances) and `oval21` (all 30). A planted bug *fires* on a benchmark when it changes at least one of its verifier’s verdicts there.

A bug is “caught by a benchmark’s own labels” when one of that benchmark’s labels shows a verdict to be wrong. VeriStressGT has a known label for every instance. An external benchmark

only has labels an outside party can establish (i.e., SAT where an attack finds a counterexample that checks out numerically). Robustness (UNSAT) cannot be certified externally, so every other instance is unlabelled, and a wrong verdict there goes unnoticed.

Benchmark	Bugs that fire	Caught by own labels	Checkable verdicts
VeriStressGT	39/42	39/42	100%
VNN-COMP <code>mnist_fc</code> (30/90 subset)	19/42	6/42	21%
VNN-COMP <code>oval21</code>	19/42	2/42	7%

- **Bugs that never fire:** 19 of the 42 bugs never change a verdict on either external benchmark, reflecting a lack of benchmark coverage. VeriStressGT exposes 16 of these 19.
- **Real bugs found:** unmodified Marabou 2.0.0 returns false SAT on the provably robust VeriStressGT instances `meap_02`, `meap_03` and `milp_s2_a`, and false UNSAT on `oval21` instance `vb_oval21_0014`, which a certified counterexample shows is SAT.
- **Subset:** the `mnist_fc` row uses 30 of 90 instances for a quicker proof of concept. More instances can only make more bugs fire, so it may understate full `mnist_fc`; `scripts/run_external_comparison.sh` runs all 90.

3 Expected Results and BAA Alignment

Every verifier in the pool gives a verdict (SAT or UNSAT) on every instance, or times out. Because each instance’s correct answer is known, the card checks each verdict and reports two metrics:

- **Scoring accuracy** (`gt_scoring_accuracy` which is the BAA metric with target $\geq 95\%$): of all SAT/UNSAT verdicts, the fraction that the labels judge correctly, i.e. a correct verdict is accepted and a wrong one is rejected.
- **Buggy verifiers caught** (`gt_detection_rate`): of the planted bugs that fired, the fraction shown to be wrong on some instance.

The run should end with `RESULT: VERIFIED`, with both metrics at 1.0 in the top-level `verdict.json`. The card passes only if all three hold:

1. the known labels score 100% of verdicts correctly;
2. they do not falsely flag a sound verifier;
3. they catch at least 75% of the bugs that fired.

Our reference run took 32 minutes on a MacBook Pro (Apple Silicon, 8 GB, CPU only) with MAGNET v0.1.0.

Scoring accuracy (2,025 SAT/UNSAT verdicts). A wrongful accusation calls a correct verdict wrong while a wrongful acquittal calls a wrong verdict correct; a tie is a majority vote with no winner.

Labels used for scoring	Correct	Wrongful accusations	Wrongful acquittals	Ties
Known labels (VeriStressGT)	2025/2025 (100%)	0	0	0
Majority vote, one buggy verifier	2015/2025 (99.5%)	0	0	10
Majority vote, full pool	1959/2025 (96.7%)	19	25	22

Buggy verifiers caught (42 planted, 39 fired)

Labels used for scoring	Caught	Sound verifiers wrongly flagged
Known labels (VeriStressGT)	39/39 (100%)	none
Majority vote, one buggy verifier	37/39 (95%)	none
Majority vote, full pool	39/39 (100%)	3: reference , ibp_only , real NeuralSAT

4 Scalability

TA2 scales the card with one parameter, **scale**: N , in the card’s **algo_params**. The runner builds a benchmark N times larger with fresh seeds (**thrust1_bench.xN/**) and runs the same verifier pool on it.

- **What grows:** N re-seeded copies of every base network, each keeping its analytic UNSAT certificate, plus N times the random and attack-hard CNN SAT instances.
- **Why results stay consistent:** every label is re-derived from construction and every SAT witness is re-certified, so expected ground-truth accuracy is 100% at any N .
- **Tested:** at **scale**: 2, 98 instances (48 UNSAT, 50 SAT); all 50 witnesses re-certified, ground-truth scoring 173/173, no control flagged (sound controls only, 9 minutes, free size-limited Gurobi license).
- **Needs:** **gurobipy** (installed by the setup script) to build larger benchmarks.

Thrust 2. A separate self-evaluation card (not the submission) predicts per-verifier timeouts: AUC is 0.77–0.92 on held-out data and 0.87–0.96 on fresh networks, all above the 0.7 target.